

5 Human serum albumin is a globular protein found in blood plasma.

Albumin has a molecular mass of 66 300 daltons.

It consists of 585 amino acids.

At pH 7.4 albumin has over 200 negative charges on the surface of each molecule.

(a) (i) Calculate the mean molecular mass of an amino acid in albumin.

Give your answer to an appropriate number of significant figures.

(1)

Answer daltons

(ii) Explain why albumin is soluble in blood plasma.

(2)

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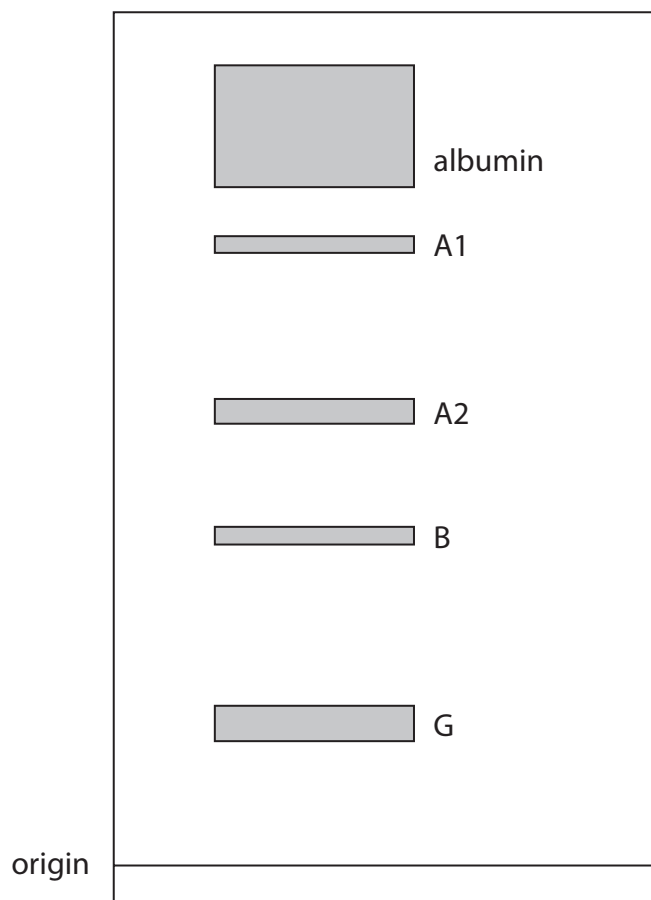
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- (b) Albumin can be separated from other proteins in blood plasma, A1, A2, B and G, by gel electrophoresis.

The diagram shows the banding pattern produced on separation of these proteins by gel electrophoresis.



- (i) Protein G has a concentration in blood plasma in the range 0.700 g dm^{-3} to 1.700 g dm^{-3} .

Albumin has a concentration in blood plasma in the range 0.525 g dm^{-3} to 1.275 g dm^{-3} .

Calculate the maximum difference in the concentration of these two proteins.

(1)

Answer g dm^{-3}

- (ii) Describe the conclusions that can be made about albumin, compared with the other four proteins present in blood plasma.

Use the information in the diagram to support your answer.

(3)

(c) Albumin binds to and activates a proton channel, hHv1.

This proton channel is present in the membranes of sperm cells. When the channel is activated, sperm cells have the ability to penetrate and fertilise egg cells.

Sperm cells are released in a liquid called seminal fluid. Seminal fluid has an albumin concentration of $15 \mu\text{mol dm}^{-3}$.

The fluid inside the uterus of the female has an albumin concentration of $500 \mu\text{mol dm}^{-3}$.

Explain the role of albumin concentration in controlling fertilisation.

(4)

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(Total for Question 5 = 11 marks)